
Web Development and Design

Lecture 07: Information Design

Information Design

- Information design addresses *how content can be organized, accessed, and managed*
- **Three key issues:**
 - **Content.** What content is available?
 - **Composition.** What views on that content do we wish to provide users?
 - **Navigation.** How do the users gain access to those views?

Information Architecture (IA)

- The overall strategy for information design usually combines both bottom-up and top-down approaches:
 - **Bottom-up:** Commonly used for small WebApps; Build pages and progressively link them into the structure.
 - **Top-down:** Commonly used for complex WebApps. It considers overall organization
 - starts with the big picture of how everything should be organized and then creating the individual pages within that structure)– the realm of the *Information Architect*.

Information Architecture (IA)

- IA acts as the **foundation** for organizing your website's content. It focuses on the **structure, labeling, and overall organization** of information.
 - During the IA process, you **plan and analyze** how users will navigate and find the information they need.

IA Characteristics

- **Focuses on User Needs:** IA prioritizes user experience by ensuring the website's content structure and organization are clear, logical, and meet user needs.
 - E.g. If you're on an e-commerce website looking for a product, good IA means you'd likely find a clear "Products" section, and within that, maybe a Product category
 - IA makes sure the **information is where you'd logically expect it to be.**
- **Hierarchical Structure:** IA establishes a hierarchy for website content, grouping related information together and defining **parent-child** relationships between pages.
- This hierarchical structure helps you narrow down your search.
 - A university website might have a **"parent" section called "Academics."** Under "Academics," there might be **"child" pages for different faculties**

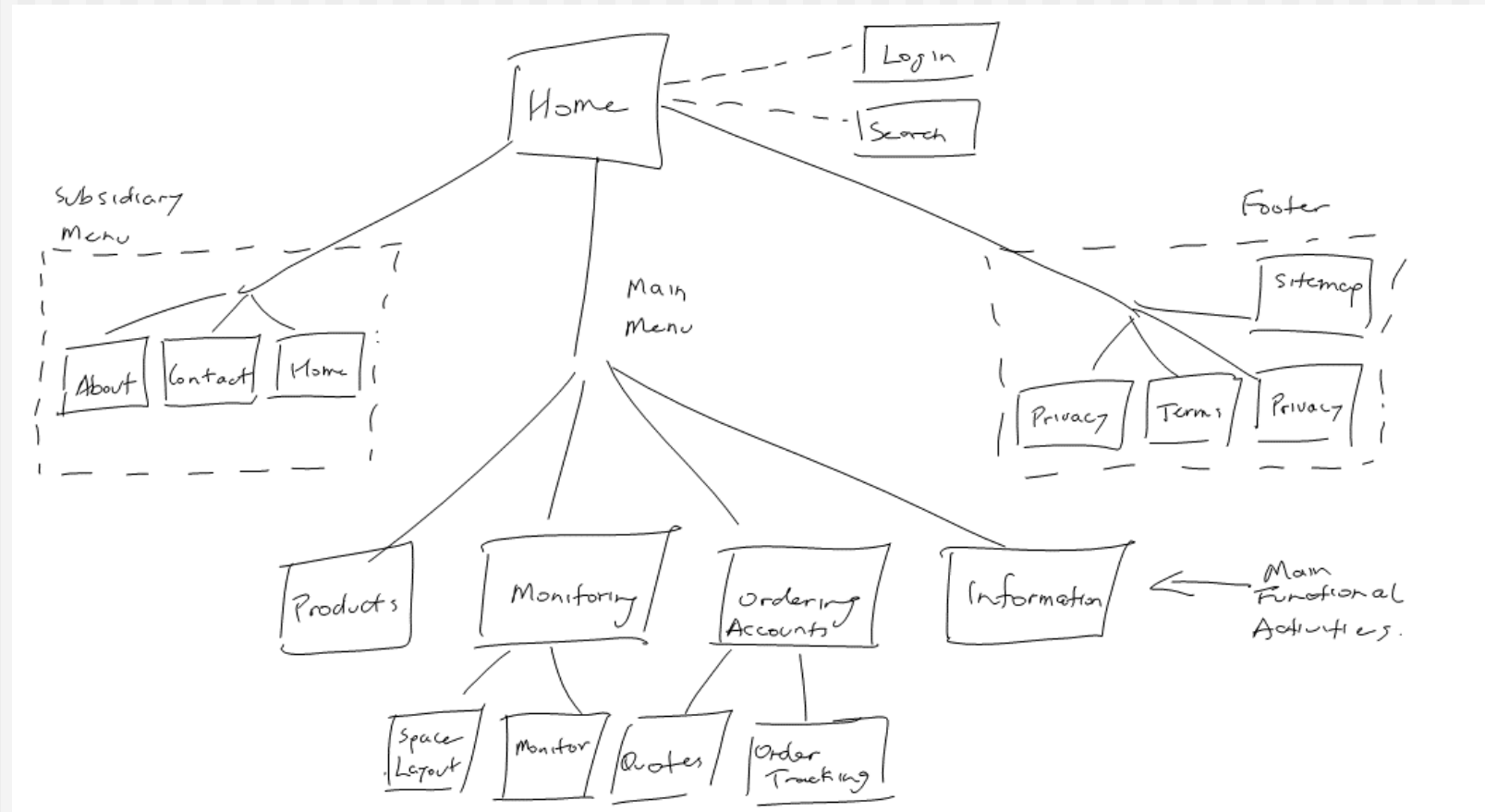
IA Characteristics

- **Temporal relations.** Certain information items may have time-based relationships, which can be only important for a certain period of time
 - e.g., a link to information about an event might only be relevant up until that event is held).
- **Usability Testing:** Effective IA involves usability testing to evaluate how users navigate the website and identify areas for improvement.
 - **E.g.** Maybe the labels are unclear, or the sections aren't organized logically, it means the IA needs improvement.
- **Flexibility and Scalability:** A good IA should be flexible enough to accommodate future growth and content addition, while still maintaining a clear structure.

Information Architecture (IA)

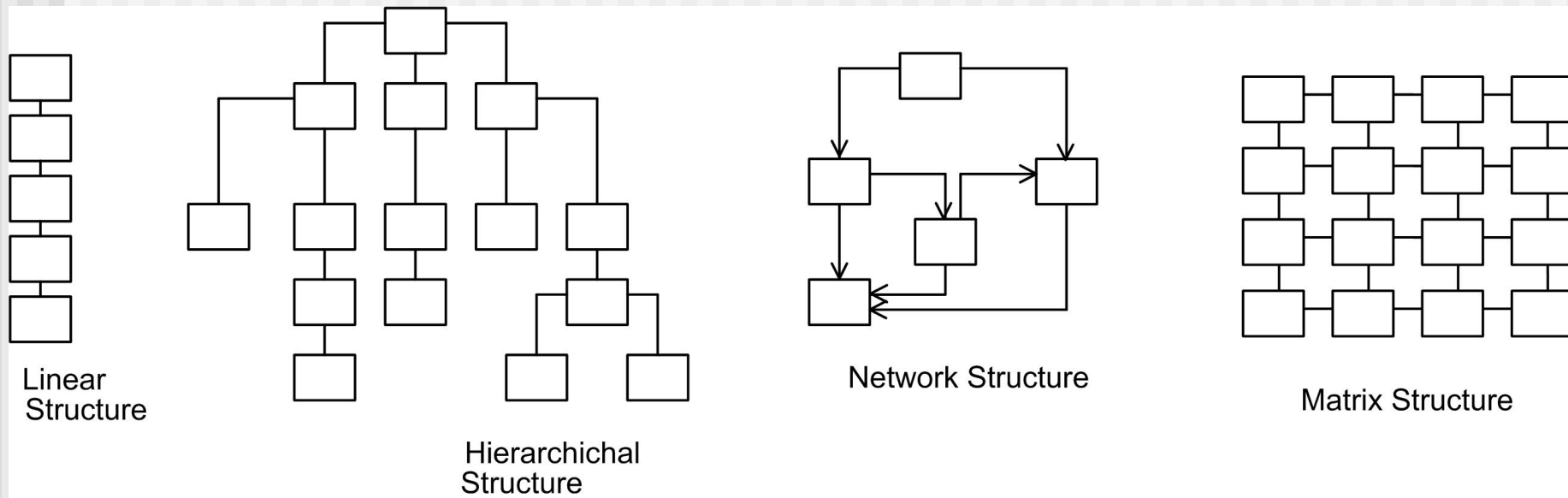
- One of the output of IA is **a *site map*** that represents the basic WebApp navigational structure.
- **A *sitemap*** is a visual representation of your website's structure. It shows the hierarchy of pages and their relationships to each other.
 - Imagine a library. Information architecture (IA) is like the overall layout and organization system of the library, deciding how books are categorized and shelved. The sitemap, on the other hand, is like a library map, visually showing the location of different sections (e.g., fiction, non-fiction) and specific bookshelves.

Example Preliminary Site Map

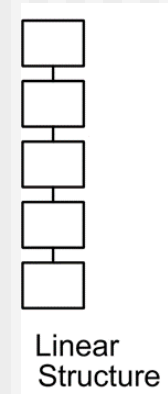


Structuring the Information Space

- The information structures that are created during information design can be classified in various ways
 - What application domains do you think are suited to each of these structures?

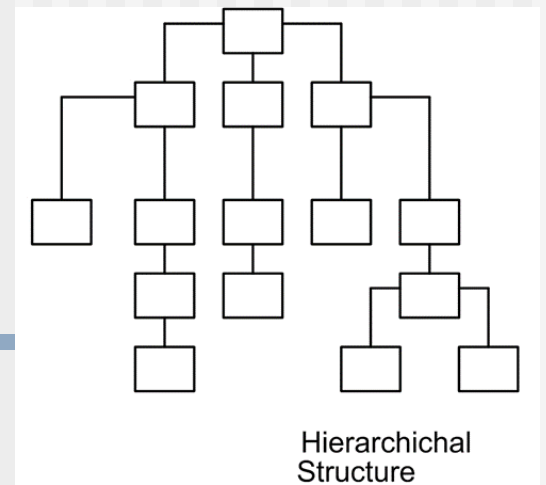


Structuring the Information Space



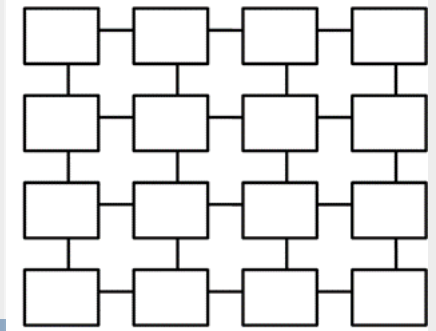
- **Linear Structure:** arranges information in **a specific order**, often chronological or based on logical steps in a process.
- Linear structures are encountered when a predictable sequence of interactions (with some variation or diversion) is common.
- A classic example might be a **tutorial presentation** in which pages of information along with related graphics, short videos, or audio are presented only after prerequisite information has been presented.
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Structuring the Information Space



- **Hierarchical structures.** This is the most common structure, organizing information in **a top-down** manner. At the very top, you have broad, general topics. As you go down, the topics become more specific and detailed.
- Examples: Website navigation menus, product catalogs

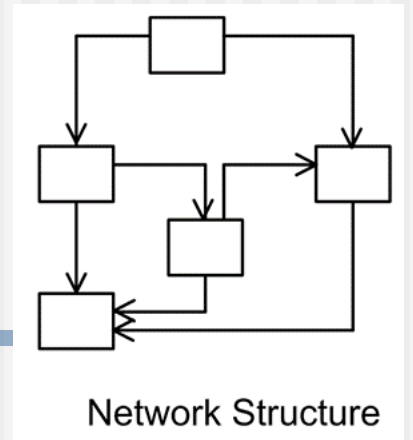
Structuring the Information Space



Matrix Structure

- **Matrix (or grid) structures.** can be applied when Web-App content **is organized categorically in two (or more) dimensions (a grid format)**, allowing for comparison and relationship identification between different categories or items. Think of a spreadsheet or comparison chart.
 - For example, consider the case in which a WebApp provides details on procedures for car repair. A matrix structure is chosen. Horizontal access to the matrix is divided into information on different problems (steering faults, ignition problems, etc.), and vertical access to the matrix is divided into information describing symptoms, causes, solutions, tools, and procedures. Hence, users might navigate the grid horizontally if they are trying to solve a particular problem and then vertically to understand the nature of that problem.

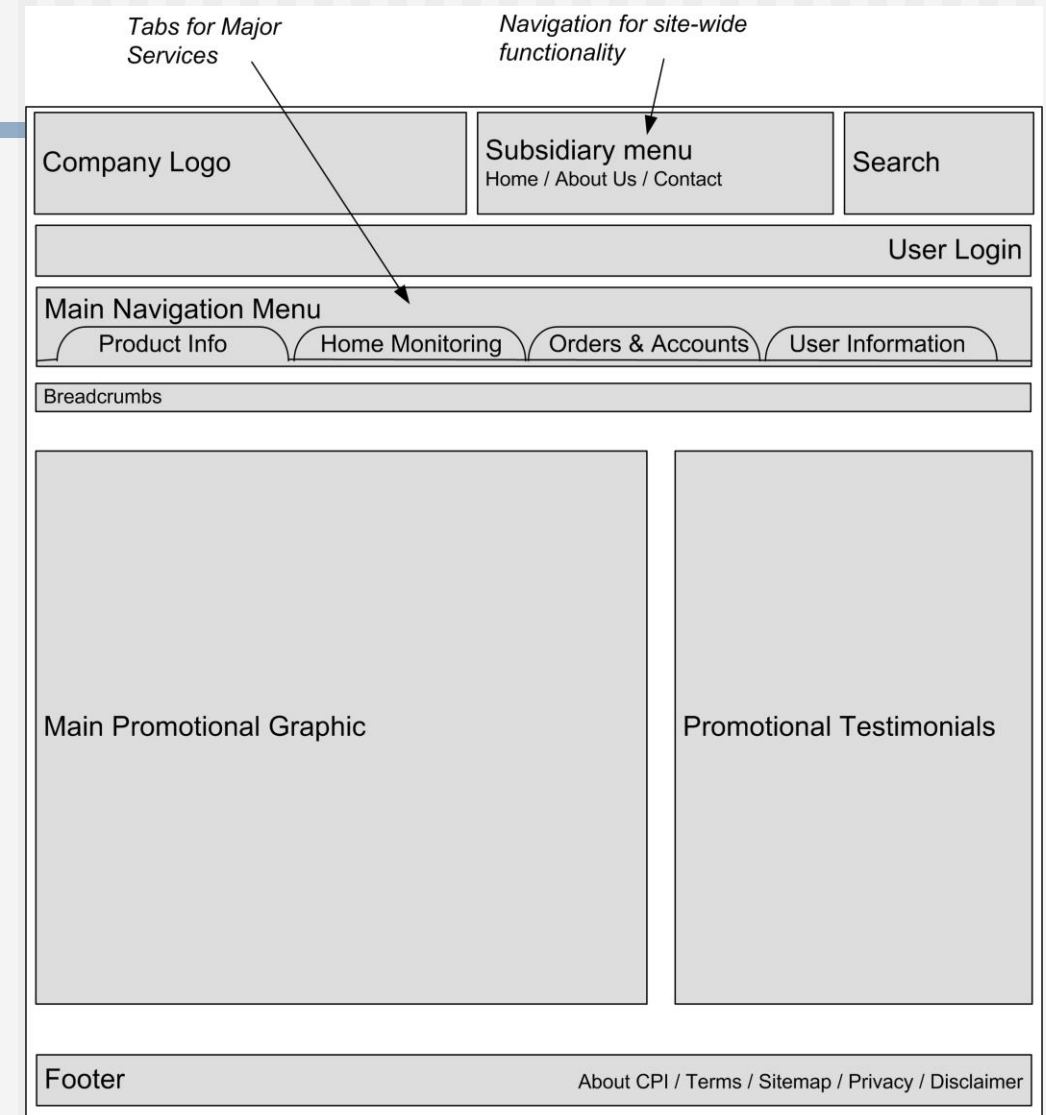
Structuring the Information Space



- **Network or graph structures.** represent information as nodes (connected points) and the relationships between them using edges (connecting lines).
- This approach can be helpful for visualizing complex interconnected information. For example, a mind map used for brainstorming ideas, the central node represents the main topic, and connecting branches represent related subtopics or keywords.
- An excellent example of a network structure is **www.wikipedia.com**. The Wikipedia content has some categorizations but is based on an incredibly rich set of cross references among the content rather than a natural hierarchy.

Wireframe Models

- A **wireframe** is a basic, **black-and-white blueprint of a webpage**, showing the important parts and how they're organized, **without worrying about colors, fonts, or images**.
- It focuses on the essential parts a webpage needs:
 - **Content Blocks:** Where the text, images, and videos will go.
 - **Navigation:** How users will move around the website (menus, buttons).
 - **Key Features:** Important functions like search bars, login areas, or product listings



Wireframe Models

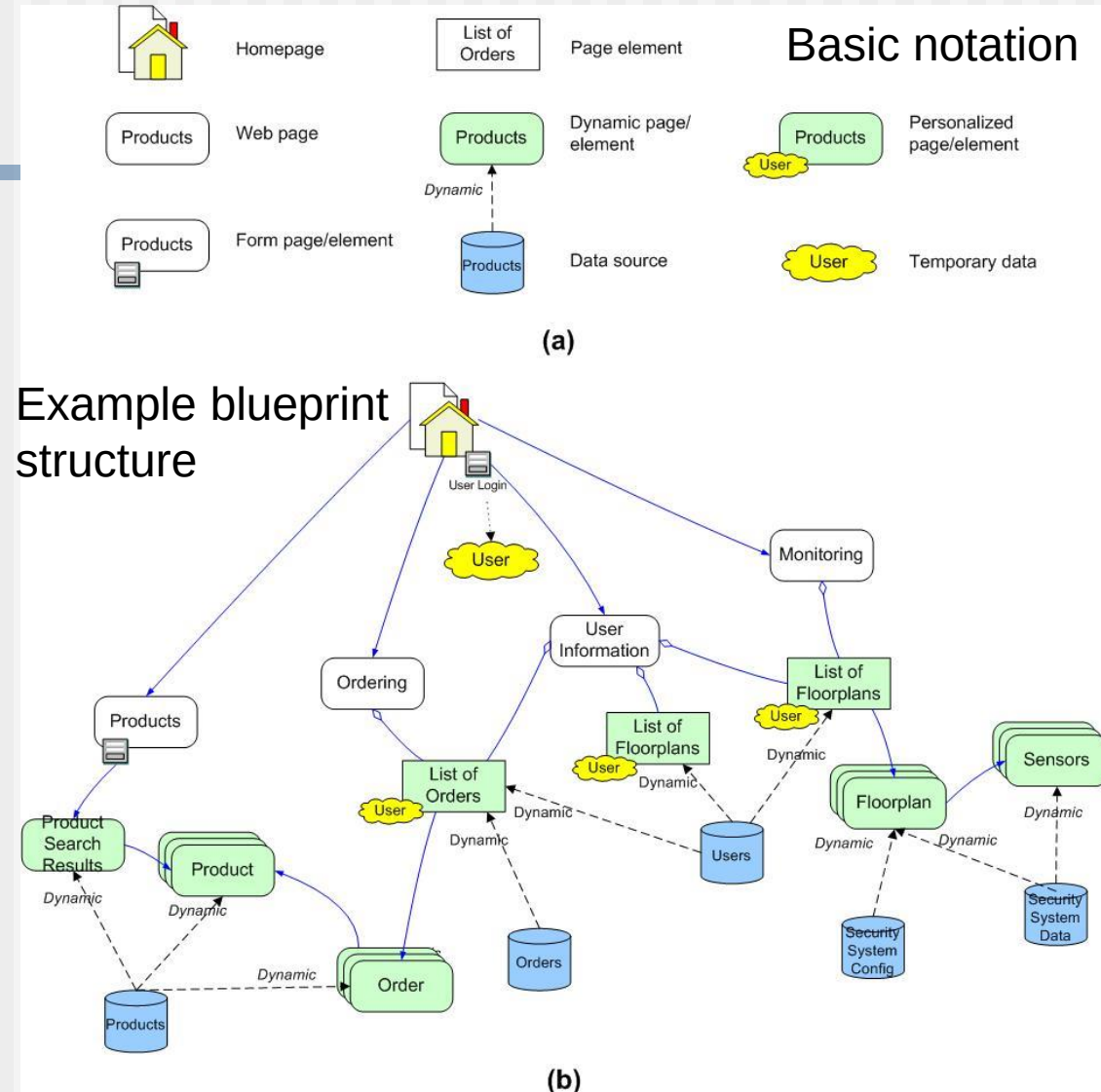
- **The wireframe is a visual plan for a webpage that helps everyone agree on the structure and how it will work before any construction begins.**

Website Blueprint

- **A website blueprint is a planning tool that can combine both textual and visual elements that** outlines the structure, content, and functionality of a website before development begins.
- A website blueprint document typically includes:
 - **Website goals and objectives:** What do you want your website to achieve?
 - **Target audience:** Who are you designing the website for?
 - **Site architecture:** How will the website be organized (sitemap)?
 - **Content plan:** What information will be included on each page?
 - **User interface (UI) design direction:** What kind of look and feel do you want for the website? (mood boards, color palettes, etc.)
 - **Technical specifications:** What technologies will be used to build the website?

Website Blueprints

- It's a high-level plan that outlines the structure and key components of the website. It focuses on the overall layout, navigation, **and how different sections of the site will connect.**
- It would heavily rely on symbols, notations, and diagrams (like flowcharts or sitemaps) to depict page layouts, navigation elements, and content organization.
- The specific format (document or diagram) can vary depending on the project's needs and the chosen tools.



Wireframe vs Blueprint

Feature	Wireframe Model	Blueprint (Website Context)
Focus	Structure & Functionality	Overall Website Plan
Level of Detail	Very Basic	More Detailed
Visual Design Elements	No (Doesn't include visual design elements like colors, fonts, or images)	May include some high-level ideas
Purpose	Early-stage planning	Website development roadmap

Understanding context

- Have you ever navigated into a complex WebApp and felt “lost in hyperspace”?
- When this happens, you lose track of where you are within (or beyond) the WebApp.
- It’s a common problem that can leave the user disoriented and unable to acquire needed information or invoke appropriate functionality.

Defining Context - Guidelines

- An information architect should be able to:
 - tune navigational support to the specific characteristics of the IA
 - design search mechanisms that lead the user to desired information while filtering out extraneous content.
 - help experienced users achieve their navigational goals more quickly
 - provide inexperienced users with additional navigational support

Defining Context - Guidelines

- **These features must be considered as part of the information design:**
 - **Clear labeling.** develop a set of local standards that lead to a clear set of labels for all link anchors.
 - Anchors describe the destination of the link and can be crucial for ensuring that users understand where they have landed when following a link
 - use action verbs (like "View," "Contact," "Learn") followed by a clear description of the destination
 - **Identity.** Each Web page should clearly identify the nature of the site or subsite to which presented information belongs
 - Without clear identity, you might feel like you've wandered off to a completely different website.

Defining Context - Guidelines

- **Breadcrumbs:** provide users with an indication of where they are and what they are currently seeing
- Breadcrumbs are typically displayed as a horizontal list of text links usually at the top of a webpage, separated by arrows (>) or other symbols. Examples:
 - **Home > Section > Sub-section > Current Page**
 - **Home > Products > Laptops**
 - **Home > Clothing > Shoes > Sneakers**
 - This example shows the user's location on an e-commerce website. They are currently viewing a specific product category (sneakers) after browsing through the general clothing and shoe categories from the homepage.

Defining Context - Guidelines

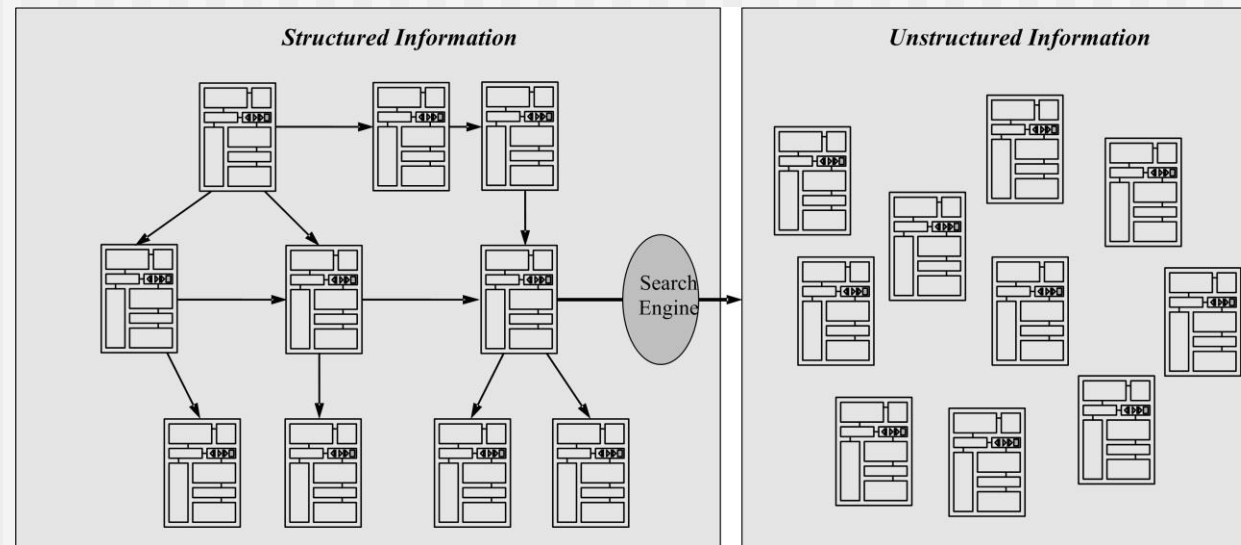
- **Global links.** are provided on every Web page and point to commonly visited WebApp locations or functions.
 - usually found in the header or footer of every single page on the website ("**Home**" - "**About Us**" - "**Contact Us**")
- **Shortcuts.** These are ways of bypassing the normal navigational route and jumping over intermediate steps straight to a particular location within the information space
 - e.g. "Recently Viewed" Section - Direct Links in Emails

Defining Context - Guidelines

- **Navigation mechanisms** that allow users to , to move through all the different pages and sections of the website (the information structure). (e.g. searching - Links within the Content - Main Navigation Menu)
 - **The search bar** is a key navigation mechanism. You type in “product name” and the website quickly shows you the pages that match your query, taking you directly to what you're looking for without needing to browse through all the product categories.

Searching Mechanisms

- In web development, searching mechanisms play a vital role in helping users find the information they need. There are two main categories of information structures to consider when designing search functionality: **structured** and **unstructured data**.



Searching Mechanisms

- **Structured Data:** refers to information that is organized in a predefined format, with clear labels and tags. Think of it like filing cabinets with folders and labels for easy organization
- **Benefits:**
 - Enables efficient and accurate search results.
 - Allows for advanced search functionalities like filtering by specific criteria.
 - Search engines can better understand the content and context of the information.
- **Examples:**
 - E-commerce websites with product details being organized into clear categories like category, price, and brand

Searching Mechanisms

■ Search Mechanisms for Structured Data:

- **Faceted Search:** Allows users to filter results based on various attributes like price range (e.g., Under 100 EGP, 100-300 EGP, Over 300 EGP), color (e.g., Blue, Red, Green, Natural) or size (commonly used in e-commerce).
- **Keyword Search with Filters:** Users enter keywords, and the search engine filters results based on pre-defined tags or categories.
 - e.g. You type "wooden box" into the search bar. The website shows you all items with "wooden" and "box" in their name or description.

Searching Mechanisms

- **Unstructured data** refers to information that lacks a predefined format and is not easily categorized.
- **Challenges for Search:**
 - Search engines have difficulty understanding the meaning and context of the information.
 - Less precise search results compared to structured data
- **Examples:**
 - Text content in website articles or blog posts.
 - Images, videos, and audio files.

Searching Mechanisms

■ Search Mechanisms for Unstructured Data:

- **Keyword Search:** Users enter keywords, and the search engine tries to match them with relevant content based on word occurrence and ranking algorithms.
- **Full-text Search:** Search engines crawl and analyze the entire text content of a website to find relevant matches.
- **Semantic Search:** More advanced techniques that try to understand the meaning and intent behind a user's search query, even if the exact keywords are not present.

Thank You